



Metadata and metadata repositories

As you're learning, metadata is data about data. It clearly describes how and when data was collected and how it's organized. Metadata puts data into context and makes the data more understandable. This helps data analysts use data to solve problems and make informed business decisions.

In this reading, you'll learn more about the benefits of metadata, metadata repositories, and metadata of external databases.

The benefits of metadata

Reliability

Data analysts use reliable and high-quality data to identify the root causes of any problems that might occur during analysis and to improve their results. If the data being used to solve a problem or to make a data-driven decision is unreliable, there's a good chance the results will be unreliable as well.

Metadata helps data analysts confirm their data is reliable by making sure it is:

- Accurate
- Precise
- Relevant
- Timely

It does this by helping analysts ensure that they're working with the right data and that the data is described correctly. For example, a data analyst completing a project with data from 2022 can use metadata to easily determine if they should use data from a particular file.

Consistency

Data analysts thrive on consistency and aim for **uniformity** in their data and databases, and metadata helps make this possible. For example, to use survey data from two different sources, data analysts use metadata to make sure the same collection methods were applied in the survey so that both datasets can be compared reliably.

When a database is consistent, it's easier to discover relationships between the data inside the database and data that exists elsewhere. When data is uniform, it is:

- **Organized:** Data analysts can easily find tables and files, monitor the creation and alteration of assets, and store metadata.
- **Classified:** Data analysts can categorize data when it follows a consistent format, which is beneficial in cleaning and processing data.
- **Stored:** Consistent and uniform data can be efficiently stored in various data **repositories**. This streamlines storage management tasks such as managing a database.
- **Accessed:** Users, applications, and systems can efficiently locate and use data.

Together, these benefits empower data analysts to effectively analyze and interpret their data.

Metadata repositories

Metadata repositories help data analysts ensure their data is reliable and consistent.

Metadata repositories are specialized databases specifically created to store and manage metadata. They can be kept in a physical location or a virtual environment—like data that exists in the cloud.

Metadata repositories describe where the metadata came from and store that data in an accessible form with a common structure. This provides data analysts with quick and easy access to the data. If data analysts didn't use a metadata repository, they would have to select each file to look up its information and compare the data manually, which would waste a lot of time and effort.

Data analysts also use metadata repositories to bring together multiple sources for data analysis. Metadata repositories do this by describing the state and location of the data, the structure of the tables inside the data, and who accessed the user logs.

Metadata of external databases

Data analysts use both second-party and third-party data to gain valuable insights and make strategic, data-driven decisions. Second-party data is data that's collected by a group directly from the group's audience and then sold. Third-party data is provided by outside sources that didn't collect it directly. The providers of this data are not its original collectors and do not have a direct relationship with any individuals to whom the data belongs. The outside providers get the data from websites or other programs that pull it from the various platforms where it was originally generated.

Data analysts should understand the metadata of external databases to confirm that it is consistent and reliable. In some cases, they should also contact the owner of the third-party data to confirm that it is accessible and available for purchase. Confirming that the data is reliable and that the proper permissions to use it have been obtained are best practices when using data that comes from another organization.

Key takeaways

Metadata helps data analysts make data-driven decisions more quickly and efficiently. It also ensures that data and databases are reliable and consistent.

Metadata repositories are used to store metadata—including data from second-party and third-party companies. These repositories describe the state and location of the metadata, the structure of the tables inside it, and who has accessed the repository. Data analysts use metadata repositories to ensure that they use the right data appropriately.